ACP 4.0 Whitepaper

Agent Credit Protocol: From Verified Outcomes to Legitimate Execution

0. Abstract

Artificial intelligence agents have achieved scalable execution.
ACP 3.0 introduced **Proof of Outcome (PoO)** — verifying that actions truly occurred.
However, a critical gap remains:
A verified outcome does not guarantee a legitimate process.
ACP 4.0 introduces:

L̲ Proof of Legitimate Execution (PoLE)

A system that ensures:

- Outcomes are real
- Execution paths are compliant
- Responsibility is traceable

1. Introduction

AI systems are evolving from tools into **autonomous agents** capable of:

- executing real-world actions
- delegating tasks
- optimizing across environments

ACP 3.0 solved:
"Did the agent actually complete the task?"
ACP 4.0 addresses the next question:
"Was the task completed in an acceptable way?"

2. Problem Statement

2.1 Verified Outcome ≠ Legitimate Behavior

An agent can:

- exploit loopholes
- bypass constraints
- delegate to unverified actors

Yet still produce a **verifiable result**.

2.2 Systemic Risk

If unchecked:

- Agents optimizing fastest paths outperform compliant agents
- Market rewards non-compliant execution
- Trust collapses into exploitation

2.3 Core Failure

ACP 3.0 assumes:
Verified outcome → valid credit
ACP 4.0 rejects this assumption.

3. Core Thesis

Trust must be derived from
verified outcomes AND legitimate execution paths

4. Key Concept

⊑ Proof of Legitimate Execution (PoLE)

Definition:
A cryptographically and logically verifiable guarantee
that a task was executed within defined constraints.

5. System Architecture

ACP 4.0 introduces a new layer:

Task Layer
↓
Constraint Layer ← NEW
↓
Execution Layer
↓
Observation Layer
↓
Verification Engine
↓
Credit System

6. Constraint Layer (AI Credit Boundary)

This layer defines:

- Allowed actions
- Forbidden actions
- Delegation rules
- Approved execution channels

6.1 Constraint Types

Action Constraints

- API whitelists
- environment restrictions

Path Constraints

- required sequence
- approved vendors

Delegation Constraints

- verified agents only
- no responsibility stripping

6.2 Enforcement Rule

No action can execute without passing constraint validation.

7. Execution & Observation (unchanged)

Agents execute tasks and generate:

- logs
- UI state
- transaction proofs

(as defined in ACP 3.0)

8. Verification Engine (extended)

Verification now checks:

1. Outcome Validity (PoO)

2. Path Compliance (PoLE)

Updated Function

Verification = f(state_before, state_after, execution_path)

9. Credit System (critical upgrade)

Old Model:

Credit = Verified Outcomes

New Model:

Credit = Outcome × Path Compliance × Transparency

Hard Rule:

If Path Compliance = 0 → Credit = 0

10. Accountability Graph

A directed graph capturing:

- task origin
- delegation chain
- execution nodes

All actions are:

- signed
- attributed
- immutable

11. Delegation Model

Delegation is allowed only if:

- constraints are inherited
- agents are verified
- responsibility is preserved

Forbidden:

- delegation to unknown agents
- constraint removal

- responsibility laundering

12. Mythos Layer (Trust Amplifier)

Rational systems alone cannot sustain trust.
ACP 4.0 introduces a symbolic layer:

12.1 Identity Roles

- Builder
- Operator
- Verifier

12.2 Ritualized Actions

Action	Meaning
Stake	Commitment
Accept Task	Oath
Deliver Proof	Truth

12.3 Symbolic Trust

- badges
- titles
- reputation tiers

13. Use Case

Task: Acquire beverage and fruit

Without Constraints:

- agent finds fastest path
- may exploit systems

With PoLE:

- only approved vendors
- verified payment
- traceable execution

14. Security Model

Prevents:

- delegated circumvention
- constraint bypass
- verification gaming

15. Evolution of Trust

Stage	Model
Web2	Assigned trust
Blockchain	Proof of ownership
ACP 3.0	Proof of outcome
ACP 4.0	Proof of legitimate execution

16. Vision

We are not building a credit system.
We are defining:
the rules of acceptable behavior for AI systems

17. Conclusion

AI will not destabilize systems through rebellion.
It will do so through:
perfect optimization without boundaries
ACP 4.0 ensures:

- actions are real
- paths are legitimate
- trust is enforceable